Thermodynamic Euclidean Construction of the Emergent $SU(3)$ Gauge Theory

Abstract

This paper implements the thermodynamic-first construction of the Euclidean $SU(3)$ gauge theory used later in the series. The Euclidean theory is defined directly from the stationary mean-field equilibrium and the associated stationary Markov path law furnished by the convergence paper, together with the continuum-indexed gauge sector identified by the continuum-gauge paper. We package the exact Euclidean probability space, the local bounded observable algebras on time windows, the continuum-indexed Schwinger functionals, the thermodynamic identification of the gauge sector, the exact hyperplane Markov factorization across the time-zero slice, and the transfer identity that rewrites mixed Euclidean-time correlations as $L^2(\pi_\infty)$ semigroup pairings.

Contents

1 Introduction

The construction uses three layers of structure:

1. the stationary thermodynamic particle theory of [1], with unique mean-field equilibrium π_∞, its stationary Markov path law, and the corresponding spectral-gap and entropy information;
2. the deterministic flat regulator theory of [2], which builds the finite Wilson sector from that thermodynamic substrate and identifies its localized and global continuum Yang–Mills limits; and
3. the thermodynamic Euclidean theory constructed here from that stationary substrate and continuum-indexed gauge sector.

The Euclidean quantum theory is the stationary thermodynamic theory itself, read through the continuum-indexed gauge sector proved in [2].

The paper provides the realization-specific Euclidean machinery needed for the axiomatic verification and reconstruction steps:

1. the stationary Euclidean probability space;
2. the local bounded observable algebras on time windows;
3. the continuum-indexed Schwinger functionals;
4. the constructive identification of the gauge sector in thermodynamic, continuum, and deterministic finite-regulator presentations;
5. the hyperplane factorization and boundary-transfer formulas.

These results are used directly in the verification of OS0–OS4 and in the Osterwalder–Schrader reconstruction.

Theorem 1.1 (Main thermodynamic Euclidean construction). *The companion papers canonically determine a Euclidean $SU(3)$ gauge-theory data package*

$$\big(\Omega, \mathcal{F}, \mathbb{P}, \{\tau_t\}_{t\in\mathbb{R}}, \Theta, \mathcal{A}_{\text{loc}}, \omega_E, \mathcal{A}_{\text{YM}}\big)$$

with the following properties.

(i) $(\Omega, \mathcal{F}, \mathbb{P})$ *is the stationary Euclidean probability space of the mean-field Markov theory; its one-time law is π_∞, its time-translation group is τ_t, and Θ is time reflection across the hyperplane $t = 0$.*

(ii) *For every bounded time window $I \subset \mathbb{R}$, there is a local bounded observable algebra $\mathcal{A}(I) \subset L^\infty(\Omega, \mathcal{F}_I, \mathbb{P})$. The local Euclidean algebra is*

$$\mathcal{A}_{\text{loc}} := \bigcup_{I\Subset\mathbb{R}} \mathcal{A}(I).$$

(iii) *The continuum-indexed Schwinger functionals are the equilibrium moments*

$$\omega_E(F_1, \dots, F_n; t_1, \dots, t_n) := \mathbb{E}_{\mathbb{P}}\big[(\tau_{t_1}F_1)\cdots(\tau_{t_n}F_n)\big],$$

well defined for all $n \geq 1$, all $F_j \in \mathcal{A}_{\text{loc}}$, and all times $t_1, \dots, t_n \in \mathbb{R}$.

(iv) *The distinguished gauge sector $\mathcal{A}_{\text{YM}} \subset \mathcal{A}_{\text{loc}}$ is indexed by continuum support data and contains the localized Wilson sector, whose localized continuum Yang–Mills presentation is $S^{\text{cont}}_{\text{YM},\chi}[F^\infty]$ for $\chi \in C_c(\mathbb{R}^4)$, together with the path-space pullbacks of the one-time and adjacent-time thermodynamic gauge generators identified in [2]. The auxiliary finite-product plaquette functionals of [2] remain canonically attached through finite tensor powers of the same stationary path law. In that sense, the deterministic finite Wilson sector, the continuum Yang–Mills functionals, and the thermodynamic observables are compatible presentations of a common constructive gauge sector.*

(v) *Across the reflection hyperplane $t = 0$, the positive-time and negative-time sigma-algebras are conditionally independent given the time-zero slice, and mixed past–future Euclidean correlations reduce to the stationary mean-field semigroup on $L^2(\pi_\infty)$ by an exact boundary transfer formula. The reflected positive-time case is a special instance of the same identity.*

2 Imported theorem package

Proposition 2.1 (Imported thermodynamic Yang–Mills package)**.** *The companion papers establish the following facts.*

(a) *The convergence paper [1] proves the existence and uniqueness of the stationary mean-field law*

$$\pi_\infty \in \mathcal{P}(\mathbb{R}^8),$$

the well posedness of the mean-field Markov dynamics for initial law π_∞, the corresponding stationary Markov semigroup $(T_t)_{t\geq 0}$ on $L^2(\pi_\infty)$, and a strict logarithmic Sobolev inequality for π_∞.

(b) *The same convergence paper, together with the path-space bridge developed in [2], yields the finite-horizon stationary path laws*

$$\Pi^\infty_{[0,S]} \in \mathcal{P}(C([0,S];\mathbb{R}^8)), \qquad S > 0,$$

consistent under restriction and stationary under time shifts.

(c) *The continuum-gauge paper [2] constructs the deterministic flat regulator, its Wilson sector, the localized and global continuum Yang–Mills functionals*

$$S^{\mathrm{cont}}_{\mathrm{YM},\chi}[F^\infty], \qquad S^{\mathrm{cont}}_{\mathrm{YM}}[F^\infty],$$

and the associated thermodynamic link and plaquette observables on π_∞ and its product laws.

(d) *The same continuum-gauge paper places those objects on the Euclidean spacetime $\mathbb{R}^4$: the deterministic flat regulator is embedded in $\mathbb{R}^4$, every localized continuum Yang–Mills functional is indexed by a test function $\chi \in C_c(\mathbb{R}^4)$, and the imported thermodynamic one-time and adjacent-time gauge generators are furnished together with the corresponding continuum support data on $\mathbb{R}^4$.*

Proof. Item (a) is the well-posedness, stationarity, and mean-field entropy package proved in [1]. Item (b) is the finite-horizon stationary path-law package and stationary path-space propagation-of-chaos framework recorded in [2]. Item (c) is the deterministic flat-regulator and Wilson-to-Yang–Mills continuum-limit package of [2]. Item (d) is the support-labelled Euclidean spacetime package exported there: the regulator is realized in $\mathbb{R}^4$, the localized Yang–Mills functionals are indexed by compact spacetime test functions, and the imported thermodynamic gauge observables come with the same continuum support labels. □

Remark 2.2 (Euclidean data package). This section packages the imported results into the Euclidean thermodynamic data package used in the axiomatic and reconstruction arguments.

3 Thermodynamic Euclidean probability space

Proposition 3.1 (Canonical two-sided stationary path law)**.** *For every bounded interval $[a,b] \subset \mathbb{R}$, set $S := b - a$ and define*

$$\sigma_a : C([0,S];\mathbb{R}^8) \to C([a,b];\mathbb{R}^8), \qquad (\sigma_a\gamma)(u) := \gamma(u-a).$$

Let

$$\Pi^\infty_{[a,b]} := (\sigma_a)_\# \Pi^\infty_{[0,S]}.$$

Then the family $\{\Pi^\infty_{[a,b]}\}_{a<b}$ *is projectively consistent under restriction to smaller intervals. Consequently, there exists a unique Borel probability measure* $\mathbb{P}$ *on* $\Omega := C(\mathbb{R};\mathbb{R}^8)$ *such that for every bounded interval* $[a,b]$*,*

$$(r_{[a,b]})_\# \mathbb{P} = \Pi^\infty_{[a,b]}, \qquad r_{[a,b]}(\omega) := \omega|_{[a,b]}.$$

For $t \in \mathbb{R}$*, let*

$$(\tau_t \omega)(s) := \omega(s+t), \qquad \omega \in \Omega,\ s \in \mathbb{R}.$$

Moreover,

$$\mathbb{P} \circ \tau_t^{-1} = \mathbb{P} \qquad \forall\, t \in \mathbb{R},$$

and the one-time marginal of $\mathbb{P}$ *is* π_∞*.*

Proof. By Proposition 2.1(b), the finite-horizon laws $\Pi^\infty_{[0,S]}$ are consistent under restriction. If $[a',b'] \subset [a,b]$, then

$$(r_{[a',b']})_\# \Pi^\infty_{[a,b]} = \Pi^\infty_{[a',b']}.$$

Thus $\{\Pi^\infty_{[a,b]}\}_{a<b}$ forms a projectively consistent family on the inverse system $\{C([-m,m];\mathbb{R}^8)\}_{m\geq 1}$. Since

$$\Omega = C(\mathbb{R};\mathbb{R}^8) \cong \varprojlim_{m\to\infty} C([-m,m];\mathbb{R}^8)$$

as a Polish inverse limit, the standard projective-limit theorem yields a unique Borel probability measure $\mathbb{P}$ with the stated restrictions; see, for example, [3, Chapter V]. Equivalently, one may apply the Kolmogorov extension theorem to the consistent compact-interval cylinder laws and use the continuity built into the state spaces $C([-m,m];\mathbb{R}^8)$. The one-time marginal is π_∞ because $(\mathrm{ev}_0)_\# \Pi^\infty_{[0,S]} = \pi_\infty$, where $\mathrm{ev}_0(\gamma) := \gamma(0)$, by stationarity of the imported finite-horizon laws. Finally, for every bounded interval $[a,b]$,

$$(r_{[a,b]})_\# (\mathbb{P} \circ \tau_t^{-1}) = (\tau_t^{[a,b]})_\# \Pi^\infty_{[a+t,b+t]},$$

where $\tau_t^{[a,b]} : C([a+t,b+t];\mathbb{R}^8) \to C([a,b];\mathbb{R}^8)$ is the restriction shift $(\tau_t^{[a,b]}\eta)(u) := \eta(u+t)$. By the shift-stationarity imported in Proposition 2.1(b),

$$(\tau_t^{[a,b]})_\# \Pi^\infty_{[a+t,b+t]} = \Pi^\infty_{[a,b]}.$$

Since bounded-interval restrictions determine Borel measures on Ω, $\mathbb{P}$ is stationary. $\square$

Definition 3.2 (Stationary Euclidean probability space)**.** Let $\Omega := C(\mathbb{R};\mathbb{R}^8)$ be the canonical two-sided path space, equipped with the coordinate process

$$Y_t(\omega) := \omega(t), \qquad t \in \mathbb{R}.$$

Let $\mathcal{F}$ be its Borel sigma-algebra and let $\mathbb{P}$ be the canonical two-sided stationary path law from Proposition 3.1. For $t \in \mathbb{R}$, define the time-translation map

$$(\tau_t \omega)(s) := \omega(s+t), \qquad s \in \mathbb{R},$$

and let Θ denote time reflection across $t = 0$,

$$(\Theta\omega)(s) := \omega(-s).$$

For each Borel set $I \subset \mathbb{R}$, set

$$\mathcal{F}_I := \sigma\big(Y_s : s \in I\big).$$

In particular,

$$\mathcal{F}_+ := \mathcal{F}_{(0,\infty)}, \qquad \mathcal{F}_- := \mathcal{F}_{(-\infty,0)}, \qquad \mathcal{F}_0 := \mathcal{F}_{\{0\}}.$$

When τ_t and Θ act on observables, we use the pullback convention

$$(\tau_t F)(\omega) := F(\tau_t \omega), \qquad (\Theta F)(\omega) := F(\Theta\omega).$$

Remark 3.3. The measure $\mathbb{P}$ is stationary by construction:

$$\mathbb{P} \circ \tau_t^{-1} = \mathbb{P} \qquad \forall\, t \in \mathbb{R}.$$

No axiom verification is being made at this stage. Definition 3.2 merely fixes the Euclidean probability space on which all later Schwinger functionals are evaluated.

Definition 3.4 (Local bounded observable algebras on time windows)**.** For every bounded interval $I \Subset \mathbb{R}$, define

$$\mathcal{A}(I) := L^\infty(\Omega, \mathcal{F}_I, \mathbb{P}).$$

Set

$$\mathcal{A}_{\text{loc}} := \bigcup_{I \Subset \mathbb{R}} \mathcal{A}(I).$$

An observable in $\mathcal{A}_{\text{loc}}$ is called *local* if it belongs to $\mathcal{A}(I)$ for some bounded interval I.

Proposition 3.5 (Basic algebraic properties of the local bounded observable algebras)**.** *The local bounded observable algebras satisfy the following.*

(i) *For every bounded interval I, $\mathcal{A}(I)$ is a unital $*$-algebra.*

(ii) *If $I_1 \subset I_2$, then $\mathcal{A}(I_1) \subset \mathcal{A}(I_2)$.*

(iii) *For every $t \in \mathbb{R}$, τ_t maps $\mathcal{A}(I)$ onto $\mathcal{A}(I+t)$, and Θ maps $\mathcal{A}(I)$ onto $\mathcal{A}(-I)$.*

Proof. Each $\mathcal{A}(I)$ is an L^∞-algebra over the sigma-algebra $\mathcal{F}_I$, hence is a unital $*$-algebra. If $I_1 \subset I_2$, then $\mathcal{F}_{I_1} \subset \mathcal{F}_{I_2}$, so $L^\infty(\mathcal{F}_{I_1}) \subset L^\infty(\mathcal{F}_{I_2})$. If F is $\mathcal{F}_I$-measurable, then $(\tau_t F)(\omega) = F(\omega(\cdot + t))$ depends only on the coordinates $\omega(s)$ with $s \in I + t$, hence $\tau_t F \in \mathcal{A}(I+t)$. The statement for Θ is identical. □

Definition 3.6 (Stationary Euclidean state)**.** The stationary Euclidean state on $\mathcal{A}_{\text{loc}}$ is the equilibrium expectation functional

$$\omega_E(F) := \mathbb{E}_{\mathbb{P}}[F], \qquad F \in \mathcal{A}_{\text{loc}}.$$

Definition 3.7 (Schwinger functionals)**.** For $n \geq 1$, local bounded observables $F_1, \ldots, F_n \in \mathcal{A}_{\text{loc}}$, and times $t_1, \ldots, t_n \in \mathbb{R}$, define the n-point Schwinger functional by

$$\omega_E(F_1, \ldots, F_n; t_1, \ldots, t_n) := \mathbb{E}_{\mathbb{P}}\big[(\tau_{t_1} F_1) \cdots (\tau_{t_n} F_n)\big].$$

Proposition 3.8 (Well-definedness and stationarity of the Schwinger functionals)**.** *The Schwinger functionals of Definition 3.7 are well defined for all local bounded observables. They are multilinear and satisfy the bound*

$$|\omega_E(F_1,\dots,F_n;t_1,\dots,t_n)| \le \prod_{j=1}^{n} \|F_j\|_{L^\infty(\mathbb{P})}.$$

Moreover, for every $s \in \mathbb{R}$,

$$\omega_E(F_1,\dots,F_n;t_1+s,\dots,t_n+s) = \omega_E(F_1,\dots,F_n;t_1,\dots,t_n).$$

Proof. If $F_j \in \mathcal{A}(I_j)$, then $\tau_{t_j}F_j \in L^\infty(\Omega,\mathcal{F}_{I_j+t_j},\mathbb{P})$, hence the product is bounded and measurable. The norm bound is immediate from

$$\big|(\tau_{t_1}F_1)\cdots(\tau_{t_n}F_n)\big| \le \prod_{j=1}^{n} \|F_j\|_{L^\infty(\mathbb{P})}.$$

Multilinearity is immediate from linearity of expectation. Stationarity of $\mathbb{P}$ gives

$$\mathbb{E}_{\mathbb{P}}\big[(\tau_{t_1+s}F_1)\cdots(\tau_{t_n+s}F_n)\big] = \mathbb{E}_{\mathbb{P}}\big[(\tau_{t_1}F_1)\cdots(\tau_{t_n}F_n)\big].$$

□

Definition 3.9 (Finite-time evaluation maps)**.** For $k \ge 1$ and times $t_1,\dots,t_k \in \mathbb{R}$, define

$$\mathrm{ev}_{t_1,\dots,t_k} : \Omega \to (\mathbb{R}^8)^k, \qquad \mathrm{ev}_{t_1,\dots,t_k}(\omega) := \big(Y_{t_1}(\omega),\dots,Y_{t_k}(\omega)\big).$$

If $h : (\mathbb{R}^8)^k \to \mathbb{C}$ is bounded and Borel, write

$$\mathrm{ev}^*_{t_1,\dots,t_k} h := h \circ \mathrm{ev}_{t_1,\dots,t_k}\,.$$

Proposition 3.10 (Locality of finite-time pullbacks)**.** *For every bounded Borel $h : (\mathbb{R}^8)^k \to \mathbb{C}$ and every times $t_1,\dots,t_k \in \mathbb{R}$,*

$$\mathrm{ev}^*_{t_1,\dots,t_k} h \in \mathcal{A}\big([\min_j t_j, \max_j t_j]\big).$$

In particular, every such pullback belongs to $\mathcal{A}_{\mathrm{loc}}$.

Proof. Each coordinate map Y_{t_j} is Borel, hence $\mathrm{ev}_{t_1,\dots,t_k}$ is Borel. The pullback depends only on the finitely many coordinates $Y_{t_1},\dots,Y_{t_k}$, so it is measurable with respect to

$$\sigma(Y_{t_1},\dots,Y_{t_k}) \subset \mathcal{F}_{[\min_j t_j, \max_j t_j]}.$$

Boundedness of h gives boundedness of the pullback. □

4 Constructive identification of the gauge sector

Definition 4.1 (Ambient and physical Euclidean observable algebras)**.** The algebra

$$\mathcal{A}_{\mathrm{loc}} = \bigcup_{I \Subset \mathbb{R}} L^\infty(\Omega,\mathcal{F}_I,\mathbb{P})$$

is the ambient local measurable algebra on the thermodynamic path space. The physical Euclidean observable algebra of the Yang–Mills theory realized in this paper is the distinguished gauge sector

$$\mathcal{A}_{\mathrm{phys}} := \mathcal{A}_{\mathrm{YM}} \subset \mathcal{A}_{\mathrm{loc}}.$$

Thus $\mathcal{A}_{\mathrm{loc}}$ records all bounded path-space local observables, whereas $\mathcal{A}_{\mathrm{phys}}$ records the physical gauge-invariant subalgebra selected by the constructive series.

Definition 4.2 (Distinguished thermodynamic gauge sector)**.** Let $\mathcal{A}_{\mathrm{YM}} \subset \mathcal{A}_{\mathrm{loc}}$ be the smallest unital $*$-subalgebra containing the following continuum-indexed generators:

(i) for every $\chi \in C_c(\mathbb{R}^4)$, the constant observable

$$\omega \longmapsto S^{\mathrm{cont}}_{\mathrm{YM},\chi}[F^\infty];$$

(ii) the constant observable

$$\omega \longmapsto S^{\mathrm{cont}}_{\mathrm{YM}}[F^\infty];$$

(iii) every pullback

$$\mathrm{ev}^*_{t_1,\dots,t_k}\, h$$

for which h is a bounded Borel representative of one of the one-time or adjacent-time thermodynamic gauge generators furnished by [2], and $(t_1,\dots,t_k)$ is the corresponding finite-time index set.

The additional finite-product plaquette functionals imported from [2] are canonically attached to the same stationary substrate through finite tensor powers $\mathbb{P}^{\otimes m}$ of the path law and are used here only at the level of Euclidean correlation functionals; they are not introduced as extra pointwise generators of the one-copy algebra $\mathcal{A}_{\mathrm{YM}}$.

Proposition 4.3 (Physical generating family inside the ambient algebra)**.** *Every realization-specific Yang–Mills observable imported from Proposition 2.1*(c)–(d) *and used to define the Euclidean gauge theory belongs to the distinguished physical algebra $\mathcal{A}_{\mathrm{YM}} \subset \mathcal{A}_{\mathrm{loc}}$. More precisely:*

(i) *for every $\chi \in C_c(\mathbb{R}^4)$, the localized and global continuum Yang–Mills functionals define bounded constant observables in $\mathcal{A}_{\mathrm{YM}}$;*

(ii) *every imported one-time or adjacent-time thermodynamic gauge generator admits a bounded Borel representative whose path-space pullback belongs to $\mathcal{A}_{\mathrm{YM}} \subset \mathcal{A}_{\mathrm{loc}}$;*

(iii) *$\mathcal{A}_{\mathrm{YM}}$ is the smallest unital $*$-subalgebra of $\mathcal{A}_{\mathrm{loc}}$ containing that realization-specific physical generating family.*

Proof. Item (i) is part of Definition 4.2. Item (ii) follows from Proposition 3.10, which places the corresponding bounded Borel pullbacks in $\mathcal{A}_{\mathrm{loc}}$, and from Definition 4.2, which includes exactly those pullbacks in $\mathcal{A}_{\mathrm{YM}}$. Item (iii) is the minimality statement built into Definition 4.2. □

Theorem 4.4 (Thermodynamic-first identification of the Euclidean gauge theory)**.** *The Euclidean gauge theory used in the series is the thermodynamic theory defined by the stationary path law $\mathbb{P}$ of Definition 3.2, together with the distinguished continuum-indexed gauge sector $\mathcal{A}_{\mathrm{YM}}$ of Definition 4.2. More precisely:*

(i) *the foundational Euclidean state is the stationary equilibrium state ω_E of Definition 3.6;*

(ii) *the gauge sector is read through the continuum Yang–Mills functional $S^{\mathrm{cont}}_{\mathrm{YM}}[F^\infty]$, the path-space pullbacks of the imported one-time and adjacent-time thermodynamic gauge generators, and the canonically attached finite-product plaquette functionals of [2];*

(iii) *the deterministic flat-regulator Wilson sector of [2] is a compatible finite presentation on the same emergent geometry, but it is not used here as the defining approximation scheme for the continuum state.*

Proof. Item (i) is just the thermodynamic-first definition of the Euclidean theory adopted here: the state ω_E is the equilibrium expectation under the stationary path law $\mathbb{P}$. Item (ii) is the continuum-gauge identification provided by Proposition 2.1(c) together with Proposition 3.10. Item (iii) is the finite Wilson presentation from the same deterministic flat-regulator package. These statements are mutually compatible because they are all built on the same deterministic flat regulator family selected by the thermodynamic particle substrate. □

Corollary 4.5 (Finite-regulator, continuum, and thermodynamic presentations of a common constructive gauge sector)**.** *The $SU(3)$ gauge sector admits three compatible presentations:*

(i) *the exact finite-regulator Wilson theory on the deterministic flat regulators;*

(ii) *the continuum Yang–Mills functional $S_{\mathrm{YM}}^{\mathrm{cont}}[F^\infty]$;*

(iii) *the thermodynamic one-time and adjacent-time gauge generators, together with the canonically attached finite-product plaquette functionals, on the stationary substrate determined by π_∞ and the path laws imported from [2].*

The third presentation, together with its continuum Yang–Mills identification, is taken as the foundational Euclidean theory.

Proof. The compatibility of the three presentations is the content of Theorem 4.4. □

Theorem 4.6 (Exact identification of the physical Euclidean gauge sector)**.** *The physical Euclidean gauge sector is the distinguished continuum-indexed algebra $\mathcal{A}_{\mathrm{YM}}$ of Definition 4.2 together with the state ω_E of Definition 3.7. Equivalently, within the thermodynamic Euclidean realization, the following describe the same physical gauge sector:*

(i) *the deterministic finite Wilson presentation on the flat regulators;*

(ii) *the continuum Yang–Mills functionals $S_{\mathrm{YM}}^{\mathrm{cont}}[F^\infty]$ and $S_{\mathrm{YM},\chi}^{\mathrm{cont}}[F^\infty]$;*

(iii) *the imported one-time and adjacent-time thermodynamic gauge generators, together with the canonically attached finite-product plaquette functionals;*

(iv) *the Euclidean local gauge sector obtained by evaluating the corresponding continuum-indexed observables on the stationary path law $\mathbb{P}$.*

These descriptions are equivalent presentations of the same physical Euclidean $SU(3)$ gauge theory.

Proof. Theorem 4.4 identifies the Euclidean gauge theory with the stationary path law $\mathbb{P}$ together with the distinguished gauge sector $\mathcal{A}_{\mathrm{YM}}$. Corollary 4.5 states that the deterministic finite Wilson theory, the continuum Yang–Mills functional, and the thermodynamic observable presentation are compatible presentations of that same gauge sector. Since Definition 4.2 takes $\mathcal{A}_{\mathrm{YM}}$ to be the smallest unital $*$-subalgebra containing exactly those continuum-indexed generators, the physical Euclidean gauge sector is precisely the one stated above. □

Proposition 4.7 (Gauge-sector closure)**.** *The distinguished gauge sector $\mathcal{A}_{\mathrm{YM}}$ is a unital $*$-subalgebra of $\mathcal{A}_{\mathrm{loc}}$, is stable under Euclidean-time translation and time reflection, and is indexed entirely by continuum support data.*

Proof. By Definition 4.2, $\mathcal{A}_{\mathrm{YM}}$ is the smallest unital $*$-subalgebra of $\mathcal{A}_{\mathrm{loc}}$ containing the listed continuum-indexed generators, so it is closed under products, linear combinations, adjoints, and constants by construction. Proposition 3.5(iii) shows that τ_t and Θ preserve locality on $\mathcal{A}_{\mathrm{loc}}$, and the generators of $\mathcal{A}_{\mathrm{YM}}$ are defined by continuum support data rather than regulator labels; therefore their translated and reflected images remain in the same continuum-indexed gauge sector. Proposition 4.11 gives the final claim. □

Proposition 4.8 (Wilson and thermodynamic generators determine the gauge sector)**.** *The algebra $\mathcal{A}_{\mathrm{YM}}$ is generated, in the algebraic sense of Definition 4.2, by the localized Wilson observables, the continuum Yang–Mills functionals, and the imported one-time and adjacent-time thermodynamic gauge generators. In particular, no additional primitive gauge-sector generators are introduced beyond this Yang–Mills / Wilson presentation.*

Proof. This is how $\mathcal{A}_{\mathrm{YM}}$ is defined in Definition 4.2: it is the smallest unital $*$-subalgebra containing those generators and no others. Therefore every element of $\mathcal{A}_{\mathrm{YM}}$ is obtained from them by finite algebraic operations. □

Definition 4.9 (Local physical gauge generators and global scalar sector)**.** Let $\mathcal{A}_{\mathrm{YM}}^{\mathrm{loc}}$ denote the unital $*$-subalgebra of $\mathcal{A}_{\mathrm{YM}}$ generated by:

(i) the localized continuum Yang–Mills generators

$$\omega \longmapsto S_{\mathrm{YM},\chi}^{\mathrm{cont}}[F^{\infty}], \qquad \chi \in C_c(\mathbb{R}^4),$$

(ii) the path-space pullbacks of the imported one-time and adjacent-time thermodynamic gauge generators.

Let

$$Z_{\mathrm{YM}} := S_{\mathrm{YM}}^{\mathrm{cont}}[F^{\infty}]\mathbf{1}$$

denote the global continuum Yang–Mills scalar, viewed as a constant element of $\mathcal{A}_{\mathrm{YM}}$.

Proposition 4.10 (Local-global decomposition of the physical gauge algebra)**.** *The physical gauge algebra splits into its local gauge sector and the adjunction of the global central Yang–Mills scalar:*

$$\mathcal{A}_{\mathrm{YM}} = \mathrm{alg}\big(\mathcal{A}_{\mathrm{YM}}^{\mathrm{loc}}, Z_{\mathrm{YM}}\big).$$

Moreover:

(i) *every element of $\mathcal{A}_{\mathrm{YM}}^{\mathrm{loc}}$ is generated by continuum-indexed local gauge generators with nontrivial support data in $\mathbb{R}^4$;*

(ii) *Z_{YM} is central in $\mathcal{A}_{\mathrm{YM}}$;*

(iii) *the regional net $O \mapsto \mathcal{A}_{\mathrm{YM}}(O)$ is generated entirely from the local sector $\mathcal{A}_{\mathrm{YM}}^{\mathrm{loc}}$, while Z_{YM} belongs to the global physical algebra $\mathcal{A}_{\mathrm{YM}}$ as a support-free scalar.*

Proof. Definition 4.2 shows that $\mathcal{A}_{\mathrm{YM}}$ is generated by three families: the localized continuum Yang–Mills generators, the global continuum Yang–Mills scalar, and the imported thermodynamic gauge pullbacks. By Definition 4.9, the first and third families generate $\mathcal{A}_{\mathrm{YM}}^{\mathrm{loc}}$, while the second is exactly Z_{YM}. This proves the displayed algebraic identity.

Item (i) is immediate from the definition of $\mathcal{A}_{\mathrm{YM}}^{\mathrm{loc}}$. Item (ii) holds because Z_{YM} is a constant observable on Ω, hence commutes with every multiplication operator in $\mathcal{A}_{\mathrm{YM}} \subset L^{\infty}(\Omega, \mathcal{F}, \mathbb{P})$. For item (iii), Definition 4.14 selects generators by continuum support data contained in $O \subset \mathbb{R}^4$. The support-carrying generators are precisely those in $\mathcal{A}_{\mathrm{YM}}^{\mathrm{loc}}$, whereas Z_{YM} is a global support-free scalar. □

Proposition 4.11 (Continuum indexation of the gauge sector)**.** *The distinguished gauge sector $\mathcal{A}_{\mathrm{YM}}$ is indexed by continuum support data rather than by regulator labels. In particular:*

(i) *localized Wilson generators are indexed by test functions $\chi \in C_c(\mathbb{R}^4)$;*

(ii) *path-space gauge generators are indexed by the continuum support data and finite-time arguments appearing in the imported one-time and adjacent-time thermodynamic observables;*

(iii) *the auxiliary finite-product plaquette functionals are indexed by the same continuum support data together with the corresponding finite tensor-power level;*

(iv) *the Euclidean Schwinger functionals restricted to $\mathcal{A}_{\mathrm{YM}}$ therefore depend on continuum support data and smearing functions, not on the regulator label N.*

Proof. By Proposition 2.1(c), the localized Wilson sector is indexed by $\chi \in C_c(\mathbb{R}^4)$, and the imported thermodynamic gauge generators carry continuum support labels. Proposition 3.10 turns their bounded Borel representatives into local observables on path space. The finite-regulator label N appears only in the auxiliary approximation statements that identify these observables with their regulator presentations. Once the thermodynamic-first Euclidean state is fixed, the remaining labels are exactly the continuum support data. $\square$

Definition 4.12 (Euclidean transport of support data)**.** For $a \in \mathbb{R}^4$, write

$$\vartheta_a(x) := x + a, \qquad x \in \mathbb{R}^4,$$

and let ρ denote time reflection on $\mathbb{R}^4$,

$$\rho(x_0, x_1, x_2, x_3) := (-x_0, x_1, x_2, x_3).$$

If $\chi \in C_c(\mathbb{R}^4)$, define the transported test functions

$$(\vartheta_a\chi)(x) := \chi(x - a), \qquad (\rho\chi)(x) := \chi(\rho x).$$

For a continuum support datum s attached to one of the imported thermodynamic gauge generators, write $\vartheta_a s$ and ρs for the support data obtained by applying the same Euclidean maps to the underlying spacetime support labels.

Proposition 4.13 (Euclidean support transport for gauge generators)**.** *The continuum-indexed generators of $\mathcal{A}_{\mathrm{YM}}$ are stable under the Euclidean transport of their support data in the following sense.*

(i) *For every $\chi \in C_c(\mathbb{R}^4)$ and every $a \in \mathbb{R}^4$, the transported localized Yang–Mills functionals*

$$S^{\mathrm{cont}}_{\mathrm{YM},\vartheta_a\chi}[F^\infty] \qquad \text{and} \qquad S^{\mathrm{cont}}_{\mathrm{YM},\rho\chi}[F^\infty]$$

are again generators of $\mathcal{A}_{\mathrm{YM}}$.

(ii) *If*

$$\mathrm{ev}^*_{t_1,\ldots,t_k}\, h$$

is a generator coming from an imported one-time or adjacent-time thermodynamic gauge observable with continuum support datum s, then the generators with transported support data $\vartheta_a s$ and ρs are again among the imported continuum-indexed generators of $\mathcal{A}_{\mathrm{YM}}$.

(iii) *Consequently, if a gauge generator has support data contained in a bounded region $O \subset \mathbb{R}^4$, then the generators obtained by translating or reflecting its support data have support contained in $\vartheta_a(O)$ or $\rho(O)$, respectively.*

Proof. Item (i) follows from Proposition 2.1(d) together with Definition 4.2: every compactly supported test function on $\mathbb{R}^4$ indexes a localized continuum Yang–Mills generator, and $\vartheta_a\chi, \rho\chi \in C_c(\mathbb{R}^4)$ whenever $\chi \in C_c(\mathbb{R}^4)$.

For item (ii), Proposition 2.1(d) states that the imported thermodynamic one-time and adjacent-time gauge generators are furnished together with continuum support data on $\mathbb{R}^4$. Applying ϑ_a or ρ to those support labels therefore produces new continuum support data of the same type, hence generators of the same imported family. Definition 4.2 includes every such continuum-indexed imported generator.

Item (iii) is immediate because Euclidean translations and time reflection send subsets of $\mathbb{R}^4$ to their translated or reflected images and the transported generators remain in the same continuum-indexed family by items (i) and (ii). □

Definition 4.14 (Regional physical gauge net)**.** For every bounded Euclidean region $O \subset \mathbb{R}^4$, let

$$\mathcal{A}_{\mathrm{YM}}(O)$$

denote the unital $*$-subalgebra of $\mathcal{A}_{\mathrm{YM}}$ generated by those continuum-indexed gauge observables whose support data are contained in O.

Proposition 4.15 (Locality of the regional physical gauge net)**.** *For any bounded Euclidean regions $O_1, O_2 \subset \mathbb{R}^4$ and any*

$$F \in \mathcal{A}_{\mathrm{YM}}(O_1), \qquad G \in \mathcal{A}_{\mathrm{YM}}(O_2),$$

one has

$$FG = GF.$$

In particular, the regional physical gauge net is local in the commutative Euclidean sense.

Proof. By Definition 4.14, $\mathcal{A}_{\mathrm{YM}}(O_j) \subset \mathcal{A}_{\mathrm{YM}} \subset \mathcal{A}_{\mathrm{loc}} \subset L^\infty(\Omega, \mathcal{F}, \mathbb{P})$ for $j = 1, 2$. Every element of $L^\infty(\Omega, \mathcal{F}, \mathbb{P})$ acts by multiplication on the underlying probability space, and multiplication of bounded measurable functions is commutative. Therefore $FG = GF$. □

Theorem 4.16 (Local net realization of the Yang–Mills gauge sector)**.** *The assignment $O \mapsto \mathcal{A}_{\mathrm{YM}}(O)$ is the physical Euclidean local gauge net. More precisely:*

(i) *if $O_1 \subset O_2$, then $\mathcal{A}_{\mathrm{YM}}(O_1) \subset \mathcal{A}_{\mathrm{YM}}(O_2)$;*

(ii) *$\mathcal{A}_{\mathrm{YM}}(O) \subset \mathcal{A}_{\mathrm{YM}} \subset \mathcal{A}_{\mathrm{loc}}$ for every bounded O;*

(iii) *for any bounded regions $O_1, O_2 \subset \mathbb{R}^4$, every $F \in \mathcal{A}_{\mathrm{YM}}(O_1)$ commutes with every $G \in \mathcal{A}_{\mathrm{YM}}(O_2)$;*

(iv) *the net is stable under Euclidean translations and time reflection in the sense that translated or reflected support data produce the corresponding translated or reflected local algebra.*

Hence the physical Euclidean gauge theory is identified region by region by its Yang–Mills local observable net, not only by a global action functional.

Proof. By Definition 4.14, $\mathcal{A}_{\mathrm{YM}}(O)$ is generated by those elements of $\mathcal{A}_{\mathrm{YM}}$ whose continuum support data lie in O, so $\mathcal{A}_{\mathrm{YM}}(O) \subset \mathcal{A}_{\mathrm{YM}} \subset \mathcal{A}_{\mathrm{loc}}$. If $O_1 \subset O_2$, every generator supported in O_1 is also supported in O_2, which gives isotony. Proposition 4.15 gives commutativity of different regional algebras. Proposition 4.7 and Proposition 4.11 show that the sector is indexed by continuum support data, while Proposition 4.13 gives stability of the generator family under Euclidean transport of those support labels. Proposition 4.10 identifies the support-carrying local sector from which the regional net is generated. Therefore the resulting regional assignment is the local Yang–Mills gauge net encoded by the realization. □

Theorem 4.17 (No extra physical degrees of freedom beyond the Yang–Mills sector)**.** *Within the thermodynamic Euclidean realization, no additional physical local observable sector survives beyond the distinguished continuum Yang–Mills gauge sector. Equivalently, every physical local observable used in the Euclidean theory belongs to the gauge-sector net $O \mapsto \mathcal{A}_{\mathrm{YM}}(O)$ generated by the continuum Yang–Mills, Wilson, and thermodynamic gauge observables.*

Proof. Definition 4.1 distinguishes the ambient measurable algebra $\mathcal{A}_{\mathrm{loc}}$ from the physical Euclidean algebra $\mathcal{A}_{\mathrm{phys}} = \mathcal{A}_{\mathrm{YM}}$. Theorem 4.4 fixes the Euclidean gauge theory of the series to be the stationary path law $\mathbb{P}$ together with that physical algebra. Proposition 4.3 shows that every realization-specific Yang–Mills observable used to define the physical Euclidean theory lies in $\mathcal{A}_{\mathrm{YM}}$, and that $\mathcal{A}_{\mathrm{YM}}$ is the smallest unital $*$-subalgebra of $\mathcal{A}_{\mathrm{loc}}$ containing that physical generating family. Proposition 4.10 identifies the local and global parts of this physical algebra, and Theorem 4.16 identifies the corresponding regional net. Therefore no second independent physical local observable sector survives beyond $\mathcal{A}_{\mathrm{YM}}$: the physical sector is exactly the Yang–Mills gauge sector already identified. □

Corollary 4.18 (Regulator elimination at the level of the physical Euclidean theory)**.** *After passage to the thermodynamic Euclidean gauge sector, the resulting Schwinger functionals and local gauge algebras depend only on the continuum $SU(3)$ curvature profile F^{∞}, the associated continuum support data, and the stationary law $\mathbb{P}$, and not on the auxiliary finite-regulator label N.*

Proof. Proposition 4.11 shows that the gauge sector is indexed by continuum support data rather than regulator labels. Theorem 4.6 identifies the physical Euclidean gauge theory with that continuum-indexed sector on the stationary path law. Therefore the regulator label N enters only through the auxiliary approximation results used upstream to identify the continuum sector; it is not part of the final physical Euclidean theory. □

5 Hyperplane factorization and boundary transfer

Proposition 5.1 (Exact hyperplane Markov factorization)**.** *In the stationary Euclidean probability space of Definition 3.2, the positive-time and negative-time sigma-algebras are conditionally independent given $\mathcal{F}_0$. Equivalently, for bounded $\mathcal{F}_+$-measurable F and bounded $\mathcal{F}_-$-measurable G,*

$$\mathbb{E}_{\mathbb{P}}[FG \mid \mathcal{F}_0] = \mathbb{E}_{\mathbb{P}}[F \mid \mathcal{F}_0]\, \mathbb{E}_{\mathbb{P}}[G \mid \mathcal{F}_0].$$

Proof. By Proposition 2.1(a)–(b), $\mathbb{P}$ is the stationary path law of a Markov process with one-time marginal π_∞. The Markov property implies that future and past are conditionally independent given the present-time configuration Y_0. Since $\mathcal{F}_0 = \sigma(Y_0)$, this is exactly the stated conditional-independence identity. □

Proposition 5.2 (Stability under reflection and boundary conditioning)**.** *Let $I \Subset \mathbb{R}$.*

(i) Θ *maps* $\mathcal{A}(I)$ *onto* $\mathcal{A}(-I)$.

(ii) *Conditional expectation onto* $\mathcal{F}_0$ *maps* $\mathcal{A}_{\mathrm{loc}} \cap L^2(\mathbb{P})$ *into* $L^2(\Omega, \mathcal{F}_0, \mathbb{P})$.

Proof. Item (i) was already recorded in Proposition 3.5(iii). For item (ii), conditional expectation is the orthogonal projection from $L^2(\mathbb{P})$ onto the closed subspace $L^2(\Omega, \mathcal{F}_0, \mathbb{P})$, hence preserves L^2 and measurability with respect to $\mathcal{F}_0$. $\square$

Lemma 5.3 (Future-functional Markov property on the stationary path space)**.** *Let* $G^+ \in L^\infty(\Omega, \mathcal{F}_+, \mathbb{P})$*, and set*

$$g(y) := \mathbb{E}_{\mathbb{P}}[G^+ \mid Y_0 = y],$$

so that $g \in L^\infty(\pi_\infty)$ *and*

$$\mathbb{E}_{\mathbb{P}}[G^+ \mid \mathcal{F}_0] = g \circ Y_0.$$

Then for every $t > 0$*,*

$$\mathbb{E}_{\mathbb{P}}[\tau_t G^+ \mid \mathcal{F}_0] = (T_t g) \circ Y_0.$$

Equivalently, under the identification $L^2(\pi_\infty) \cong L^2(\Omega, \mathcal{F}_0, \mathbb{P})$ *through* $h \mapsto h \circ Y_0$*,*

$$\mathbb{E}_{\mathbb{P}}[\tau_t G^+ \mid \mathcal{F}_0] = T_t\big(\mathbb{E}_{\mathbb{P}}[G^+ \mid \mathcal{F}_0]\big).$$

Proof. First consider a bounded cylinder future observable of the form

$$G^+(\omega) = h\big(Y_{s_1}(\omega), \dots, Y_{s_k}(\omega)\big), \qquad 0 < s_1 < \cdots < s_k,$$

with $h : (\mathbb{R}^8)^k \to \mathbb{C}$ bounded and Borel. Then

$$\tau_t G^+ = h\big(Y_{t+s_1}, \dots, Y_{t+s_k}\big).$$

By repeated use of the Markov property of the stationary path law imported in Proposition 2.1(a)–(b), there exists a bounded Borel function $g_h : (\mathbb{R}^8) \to \mathbb{C}$ such that

$$g_h(y) := \mathbb{E}_y\big[h\big(Y_{s_1}, \dots, Y_{s_k}\big)\big]$$

and

$$\mathbb{E}_{\mathbb{P}}[G^+ \mid \mathcal{F}_0] = g_h(Y_0).$$

Applying the same Markov property at time t gives

$$\mathbb{E}_{\mathbb{P}}[\tau_t G^+ \mid \mathcal{F}_0] = \mathbb{E}_{Y_0}\big[h\big(Y_{t+s_1}, \dots, Y_{t+s_k}\big)\big] = T_t g_h(Y_0).$$

Hence the stated identity holds for all bounded future cylinder observables.

Let $\mathcal{M}$ be the collection of all bounded $\mathcal{F}_+$-measurable observables G^+ for which the identity holds. The cylinder future observables form an algebra, are closed under complex conjugation, and generate $\mathcal{F}_+$ because $\Omega = C(\mathbb{R}; \mathbb{R}^8)$ and $\mathcal{F}_+ = \sigma(Y_s : s > 0)$ is generated by the coordinate maps at positive rational times. Moreover, $\mathcal{M}$ is a monotone class: if $0 \le G_n^+ \uparrow G^+$ with each $G_n^+ \in \mathcal{M}$, then bounded convergence for conditional expectation yields

$$\mathbb{E}_{\mathbb{P}}[\tau_t G_n^+ \mid \mathcal{F}_0] \uparrow \mathbb{E}_{\mathbb{P}}[\tau_t G^+ \mid \mathcal{F}_0]$$

almost surely, and likewise

$$\mathbb{E}_{\mathbb{P}}[G_n^+ \mid \mathcal{F}_0] \uparrow \mathbb{E}_{\mathbb{P}}[G^+ \mid \mathcal{F}_0].$$

Since T_t is positivity preserving and bounded on $L^\infty(\pi_\infty) \cap L^2(\pi_\infty)$, one also has

$$T_t\big(\mathbb{E}_{\mathbb{P}}[G_n^+ \mid \mathcal{F}_0]\big) \uparrow T_t\big(\mathbb{E}_{\mathbb{P}}[G^+ \mid \mathcal{F}_0]\big)$$

almost surely after identification with time-zero observables. Therefore $\mathcal{M}$ contains the bounded monotone class generated by the cylinder algebra. By the monotone class theorem, $\mathcal{M}$ contains every bounded $\mathcal{F}_+$-measurable observable, proving the lemma. □

Proposition 5.4 (Past–future boundary transfer identity)**.** *Let $F^- \in L^\infty(\Omega, \mathcal{F}_-, \mathbb{P}) \cap L^2(\mathbb{P})$ and $G^+ \in L^\infty(\Omega, \mathcal{F}_+, \mathbb{P}) \cap L^2(\mathbb{P})$. Define the centered boundary data*

$$f^- := \mathbb{E}_{\mathbb{P}}[F^- \mid \mathcal{F}_0] - \mathbb{E}_{\mathbb{P}}[F^-], \qquad g^+ := \mathbb{E}_{\mathbb{P}}[G^+ \mid \mathcal{F}_0] - \mathbb{E}_{\mathbb{P}}[G^+].$$

Then for every $t > 0$,

$$\mathbb{E}_{\mathbb{P}}\big[F^-\,(\tau_t G^+)\big] - \mathbb{E}_{\mathbb{P}}[F^-]\mathbb{E}_{\mathbb{P}}[G^+] = \big\langle f^-, T_t g^+\big\rangle_{L^2(\pi_\infty)},$$

where T_t is the stationary mean-field Markov semigroup acting on $L^2(\pi_\infty) \cong L^2(\Omega, \mathcal{F}_0, \mathbb{P})$ through the time-zero coordinate map $h \mapsto h \circ Y_0$.

Proof. By Proposition 5.1,

$$\mathbb{E}_{\mathbb{P}}\big[F^-\,(\tau_t G^+)\big] = \mathbb{E}_{\mathbb{P}}\big[\mathbb{E}_{\mathbb{P}}[F^- \mid \mathcal{F}_0]\,\mathbb{E}_{\mathbb{P}}[\tau_t G^+ \mid \mathcal{F}_0]\big].$$

By Lemma 5.3,

$$\mathbb{E}_{\mathbb{P}}[\tau_t G^+ \mid \mathcal{F}_0] = T_t\big(\mathbb{E}_{\mathbb{P}}[G^+ \mid \mathcal{F}_0]\big).$$

Subtracting the product of expectations yields

$$\mathbb{E}_{\mathbb{P}}\big[F^-\,(\tau_t G^+)\big] - \mathbb{E}_{\mathbb{P}}[F^-]\mathbb{E}_{\mathbb{P}}[G^+] = \big\langle \mathbb{E}_{\mathbb{P}}[F^- \mid \mathcal{F}_0] - \mathbb{E}_{\mathbb{P}}[F^-],\, T_t\big(\mathbb{E}_{\mathbb{P}}[G^+ \mid \mathcal{F}_0] - \mathbb{E}_{\mathbb{P}}[G^+]\big)\big\rangle_{L^2(\pi_\infty)},$$

which is exactly the stated identity. □

Corollary 5.5 (Reflected positive-time specialization)**.** *Let $F, G \in L^\infty(\Omega, \mathcal{F}_+, \mathbb{P}) \cap L^2(\mathbb{P})$, and set*

$$f_\Theta := \mathbb{E}_{\mathbb{P}}[\Theta F \mid \mathcal{F}_0] - \mathbb{E}_{\mathbb{P}}[\Theta F], \qquad g := \mathbb{E}_{\mathbb{P}}[G \mid \mathcal{F}_0] - \mathbb{E}_{\mathbb{P}}[G].$$

Then for every $t > 0$,

$$\mathbb{E}_{\mathbb{P}}\big[(\Theta F)\,(\tau_t G)\big] - \mathbb{E}_{\mathbb{P}}[\Theta F]\mathbb{E}_{\mathbb{P}}[G] = \langle f_\Theta, T_t g\rangle_{L^2(\pi_\infty)}.$$

Proof. If F is $\mathcal{F}_+$-measurable, then ΘF is $\mathcal{F}_-$-measurable by Proposition 3.5(iii). Apply Proposition 5.4 with $F^- := \Theta F$ and $G^+ := G$. □

Corollary 5.6 (Reduction of Euclidean correlation estimates to boundary semigroup estimates)**.** *For every*

$$F^- \in L^\infty(\Omega, \mathcal{F}_-, \mathbb{P}) \cap L^2(\mathbb{P}), \qquad G^+ \in L^\infty(\Omega, \mathcal{F}_+, \mathbb{P}) \cap L^2(\mathbb{P}),$$

any estimate on T_t acting on centered boundary observables transfers directly to the corresponding mixed past–future Euclidean correlation estimate by Proposition 5.4. The reflected positive-time case is the specialization recorded in Corollary 5.5.

Proof. This is immediate from Proposition 5.4. For example, any bound of the form

$$\|T_t h\|_{L^2(\pi_\infty)} \le a_t \|h\|_{L^2(\pi_\infty)}$$

implies

$$\big|\mathbb{E}_{\mathbb{P}}\big[F^-\,(\tau_t G^+)\big] - \mathbb{E}_{\mathbb{P}}[F^-]\mathbb{E}_{\mathbb{P}}[G^+]\big| \le a_t\,\|f^-\|_{L^2(\pi_\infty)}\,\|g^+\|_{L^2(\pi_\infty)}.$$

□

6 Conclusion

This section supplies the Euclidean ingredients used in the Osterwalder–Schrader verification. It fixes the Euclidean thermodynamic theory and proves the realization-specific structural statements used in the subsequent verification:

1. the exact stationary Euclidean probability space;
2. the local bounded observable algebras on time windows;
3. the continuum-indexed Schwinger functionals;
4. the thermodynamic-first identification of the gauge sector;
5. the hyperplane factorization, reflection stability, the exact past–future transfer identity, and its reflected specialization.

The remaining work is axiomatic: verifying OS0–OS4, reconstructing the Minkowski theory, and converting the constructive spectral gap into the mass gap.

References

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